Array 28 (2025) 100533

Contents lists available at ScienceDirect

Array

journal homepage: www.elsevier.com/locate/array

Deep reinforcement learning-based access class barring for mMTC access control in cellular networks

Workineh Gebeye Abera [a,*], Thomas O. Olwal [b], Yihenew Wondie Marye [a]

[a] School of Electrical and Computer Engineering, Addis Ababa Institute of Technology, Addis Ababa University, King George VI Street, Addis Ababa, Ethiopia
[b] Department of Electrical Engineering/FSATI, Tshwane University of Technology, Pretoria 001, South Africa

ARTICLE INFO

Keywords:
Massive machine type communication
LTE-A
5G NR
Access control
Access class barring
Deep reinforcement learning

ABSTRACT

Internet-of-things (IoT) is generating great income to countries' economy by interconnecting billions of Users Equipment(UEs). Massive Machine Type Communication (mMTC), a use case of IoT, enables this huge number of UEs interconnections using cellular networks (LTE-A and 5G NR). However, when sporadic large number of M2M UEs try access the eNB or gNB, which primarily supports H2H communication, there will be random access channel congestion. Access Class Barring (ACB) is an access control scheme to reduce the congestion. But, the static ACB scheme does not give the best access success probability performance in such dynamic environment. As a result, we proposed a mechanism which dynamically adjusts the barring factor parameter of the scheme based on the arrival traffic using deep reinforcement learning to improve its access success performance. The expermental results show that the mechanism achieves maximum access success probability (0.9998), meeting the theoretical optimum ACB performance ($P_s \geq 0.95$). In addition, for number of M2M UEs $N_M \leq 30,000$, the mechanism achieves similar access success (99.98%) as a RL-based ACB scheme, but with lower access delay and energy consumption; and under high traffic load ($N_M > 30,000$), the mechanism achieves better access success than the RL-based ACB scheme.

1. Introduction

The emerging Internet of Things (IoT) is required to connect everything, machine-type and human-type users. Consequently, the global IoT connection is expected to reach 40 billion in 2033 [1–3], which is anticipated to generate huge income (934 billion USD) [4]. One of the use cases of IoT is massive Machine Type Communication (mMTC),[1] which enables the Machine-to-Machine (M2M) interconnection. mMTC is applied in asset tracking, agricultural mechanization access services, smart metering, wastewater network monitoring, etc. [5,6]. Hence, mMTC is such highly applicable and emerging technology that draws research attention. mMTC preferably uses cellular networks (LTE-A, 5G NR and beyond) to interconnect the M2M Users Equipment (UEs) for its ubiquitous connectivity, security, mobility and roaming support, high Signal-to-Interference plus Noise Ratio (SINR), etc. [6–9]. In addition, the deployment cost is low when we use the existing cellular network infrastructure [10].

When a User Equipment (UE) wants to transmit data to the cellular network, it has to request the base station (eNB or gNB)[2] for access by sending an access request message called preamble through Physical Random Access Channel (PRACH), a resource with specified time–frequency, detailed in Section 3. The preambles, signatures generated by the base station, per cell are limited in number. Each UE randomly picks a preamble from these preambles to signal its access request in a RACH. When large number of UEs synchronously request the base station for access over the PRACH duration, two or more UEs may randomly pick the same preamble, which results in preamble collision. This in turn leads to access failure. When the number of preamble collisions is large and there is little or no successful access, Random Access Channel (RACH) congestion has occurred. Such a problem typically occurs when massive number of M2M UEs tend to access the base station during power recovery, scheduled access, etc. The RACH congestion causes intolerable access delay, packet loss, and even service unavailability.

Therefore, RACH congestion in cellular networks supporting mMTC, in addition to H2H communication, is a severe challenge that draws a lot of research attention. A possible solution is to limit the number of access requests per RACH to the number of available preambles

* Corresponding author.
E-mail addresses: workineh.gebeye@aait.edu.et, workineh.gebeye.abera@gmail.com (W.G. Abera), olwalto@tut.ac.za (T.O. Olwal), yihenew.wondie@aait.edu.et (Y.W. Marye).

[1] Note that we may use the words machine-type communication and machine-to-machine communication interchangeably.
[2] Note that base station refers to eNB or gNB).

https://doi.org/10.1016/j.array.2025.100533
Received 7 March 2025; Received in revised form 1 September 2025; Accepted 8 October 2025
Available online 17 October 2025
2590-0056/© 2025 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

per RACH. One of such methods is the Access Class Barring (ACB) scheme, introduced by 3GPP, detailed in Section 3. The scheme controls the RACH congestion using its two parameters namely, barring factor (P_{ACB}) and barring time (T_{ACB}), both generated by the base station. The barring factor determines whether a UE sends an access request or not and the barring time determines the time a UE has to wait before retransmitting a preamble after access failure. For optimal pair of parameters configuration, the scheme is shown to be effective in controlling RACH congestion with optimal access success probability (P_s) of $P_s \geq 0.95$ [11]. However, the scheme does not give the best access success probability performance by using *static* pair of parameters, as shown in [12] for optimal pair of parameters ($P_{ACB} = 0.5, T_{ACB} = 4$ s), in dynamic cellular network where H2H and sporadic M2M UEs coexist. Therefore, a possible solution is to dynamically adjust the barring factor based on the variation in the access request traffic intensity. This can give a barring factor value appropriate to the number of upcoming access requests, as detailed in Section 3.

Accordingly, in this paper we propose a mechanism which dynamically adjusts the barring factor according to the access request traffic intensity in a cellular network in which H2H and massive M2M UEs coexist. To adjust the barring factor the mechanism uses deep reinforcement learning algorithm, namely Dueling Double Deep Q-Learning (Dueling DDQL). We strictly used 3GPP standards in our system implementation, and hence our mechanism conforms to the specifications of the access procedure in cellular networks (LTE-A and 5G NR), enabling seamless integration into the networks. To be specific, our dynamic access procedure is designed based on LTE-A Release 14 random access procedure specifications. The motivation behind using the LTE-A model arises from our ambition to use the LTE-A random access resources, primarily designed for H2H random access, for both M2M and H2H access. This idea works equally well for 5G NR as LTE-A bands are included in Frequency Range 1 (FR1) (0.45 GHz–6 GHz) band of 5G NR and M2M communication is suitably supported in this band but not in Frequency Range 2 (FR2) (24 GHz–52.6 GHz). As a result, the mechanism can be integrated into the LTE-A network as well as 5G NR network in the FR1 band. This gives additional capacity for the eNB or gNB which also uses LTE-M and NB-IoT to support the M2M UEs random access procedure [13]. Therefore, the main contributions of the article are as follows:

1. To implement a dueling double deep Q-learning-based ACB mechanism which dynamically adjusts the *barring factor* in response to the dynamic *access request traffic intensity* to improve the access success probability performance of the ACB scheme.
2. To evaluate the access success performance of the proposed mechanism under different M2M UE traffic loads which follow the 3GPP defined traffic model, while using real records of an operator for the H2H traffic model.
3. To compare the performance of the proposed mechanism and a RL-based ACB scheme in [12] to show that our mechanism ensures the UEs maximum access success with lower access delay and energy consumption.

The rest of the paper is organized as follows. Section 2 reviews works related to our proposal. The system background concepts, random access procedure and ACB scheme, are explained in Section 3. Section 4 details the system model analysis and implementation of the proposed mechanism. The simulation results and the associated discussions are included in Section 5. Finally, in Section 6 we present our concluding remarks.

2. Related works

The (four-step) contention-based random access procedure is widely adopted and effective access control method for H2H UEs in LTE-A and 5G NR networks. However, its performance degrades largely when it is used for M2M UEs (with massive and sporadic access requests) access control. Hence, a number of optimization methods have been proposed to enable the basic random access procedure cope up with massive access requests. One of such optimization methods which draws great research attention is the ACB scheme, which is proposed by 3GPP. The ACB scheme deliberately delays the starting time of some of the access requests and is effective in reducing RACH congestion with optimal access success probability (P_s) performance $P_s \geq 0.95$ for optimal parameter configurations [11]. However, in [11] it is shown that the scheme does not give the best access success probability for static pair of optimal parameters in a cellular network in which M2M and H2H UEs coexist. A possible method is to dynamically adjust the barring factor parameter based on the intensity of arriving access request. Accordingly, in the past years a number of proposals which dynamically tune the barring factor using analytical methods have been introduced [14–16]. However, such approaches are not suitable for current and future cellular networks with large number of UEs and complex data processing. The current trend is to use machine learning algorithms, mostly Reinforcement Learning (RL), for the same purpose [17]. RL-based access control methods can easily learn from the environment to determine the barring factor value appropriate to the traffic intensity in the upcoming RAO.

Accordingly, a number of dynamic ACB schemes in the literature use machine learning techniques to tune the barring factor based on the dynamic arrival traffic. The authors in [18] proposed a strategy that adaptively adjusts the barring factor based on the number of devices accessing the gNB to reduce the access delay. The number of accessing devices is estimated using discrete generalized pursuit algorithm, which has weaker generalization ability than RL algorithms. Though the algorithm is shown to achieve similar performance with the theoretical optimal ACB scheme, the result is based on ideal scenario where only M2M UEs compete for RACH resources and hence could be different in the presence of H2H UEs. Similarly, in [19], a Long Short Term Memory (LSTM)-based ACB scheme is used to adjust the barring factor based on the upcoming (traffic load) number of preamble transmissions, which is predicted using LSTM neural network by considering the number of preamble transmissions as a time series data. The proposed scheme achieves a high success rate without significantly increasing the access delay and the number of access attempts. In addition, in [20] a model-based approach is used to obtain the optimal barring factor using the estimates of the number of access requests. To do so, the authors first derived the analytical expression for the optimal baring factor as standardized by 3GPP. Then, the type and number of accessing devices, estimated from measurements acquired by the base station by using neural networks, are plugged into the analytical expression to obtain the optimal barring factor. The method is shown to have a performance close to the theoretical optimum but does not show the effect of M2M UEs on H2H UEs access, as only M2M UEs traffic is considered. The above methods in [18–20] adjust the barring factor based on the estimate of the traffic load in the upcoming RAO and hence whether the adjusted barring factor is appropriate to the upcoming traffic load depends on the accuracy of the traffic load estimation method. In our method, the barring factor is adjusted based on the reward obtained as a result of the direct action (broadcasting a barring factor) taken by the base station. In [21], a multi-rate ACB algorithm is proposed to tune the barring rates of different service classes using reinforcement learning algorithm. The algorithm is implemented and evaluated following the 3GPP extended ACB scheme specifications. The method is shown to maximize the RACH utilization but for a scenario where only M2M UEs exist. In [12], Q-learning algorithm is used to dynamically adjust the barring factor in a practical scenario where M2M and H2H UEs coexist. The scheme is shown to have better access success probability and worse access delay performances than the static optimal ACB scheme ($P_{ACB} = 0.5, T_{ACB} = 4$ s). However, the barring factor adaptation accuracy is limited to the Q-learning algorithm's overestimation problem

and limited generalization ability. In addition, the performance of the scheme beyond 30,000 M2M UEs is uncertain.

In this paper, dueling double deep Q-learning algorithm is used to dynamically adjust the barring factor parameter according to the changing access request traffic. Our aim is to have a better access success probability performance than the static optimal ACB scheme. In addition, the mechanism is required to have better access success probability performance than our reference work [12] under high traffic load (above 30,000 M2M UEs) conditions.

3. System background

This section describes the access procedure that a UE follows to establish connection with the cellular network (LTE-A or 5G NR). It also explains the ACB scheme.

3.1. Random access procedure and control

When a UE wants to transmit data over the cellular network, it has to establish a connection with the base station. To achieve this the UE sends an access request message to the base station through the carrier of PRACH. The message is randomly selected from a pool of signatures called preambles, which are generated by the base station from Zadoff–Chu sequence [11,22]. There are up to 64 orthogonal preambles per cell. Therefore, sending an access request message is the first step of the access process called random access procedure. Users equipment initiate it during initial access to the cellular network, transition from idle to connected state, handover, and response to base station's paging message.

The PRACH is an uplink channel that is used to carry the preamble to the base station during access procedure. The PRACH is a time-frequency Resource Block (RB) which is 1 ms wide in time 6 RBs wide in frequency domain. Thus, a PRACH has a bandwidth of $6 * 180$ kHz = 1.08 MHz [23,24]. The PRACHs occur in a regular time interval determined by *prach-ConfigIndex* parameter which is broadcasted by the base station in a system information block. These regularly spaced PRACHs are specifically named as Random Access Opportunities (RAOs) [11]. A UE undergoing a random access procedure begins by sending its access request message called *preamble* to the base station. Preambles are a set of sequences generated from Zadoff–Chu sequence and there are up to 64 orthogonal preambles per cell [11,22].

In LTE-A or 5G systems, there are two modes of random access procedure, contention-free and contention-based. In contention-free mode, preambles are assigned to a UE by the base station. Hence, simultaneous transmission of the same preamble by different UEs within a RAO does not occur. This provides low latency service for services with high priority, like the cases of handover and positioning. Whereas, in the contention-based mode, which is used for regular communication services, a UE randomly selects a preamble from a set of available preambles through competition among other UEs. In this case, multiple UEs may select the same preamble in the same RAO, which may lead to preamble collision, resulting in access failure. However, the number of possible preamble collisions can be minimized through proper access control method. This is where the need for our proposed method comes from.

The contention-based random access procedure involves the exchange of messages between the UE and the base station in a number of steps equal to the number of messages exchanged. Hence, currently there are two types of it, namely *four-step* random access procedure (included in 3GPP Release 13 and 14) and *two-step* random access procedure (included in 3GPP Release 16) [25–27]. For explanation simplicity and historical precedence, let us first discuss the four-step random access procedure, followed by the two-step random access procedure, and finally a remark on our choice among the two.

The *four-step* random access procedure involves the exchange of signaling messages between the UE and the base station in four steps as shown in Fig. 1 The steps are as follows:

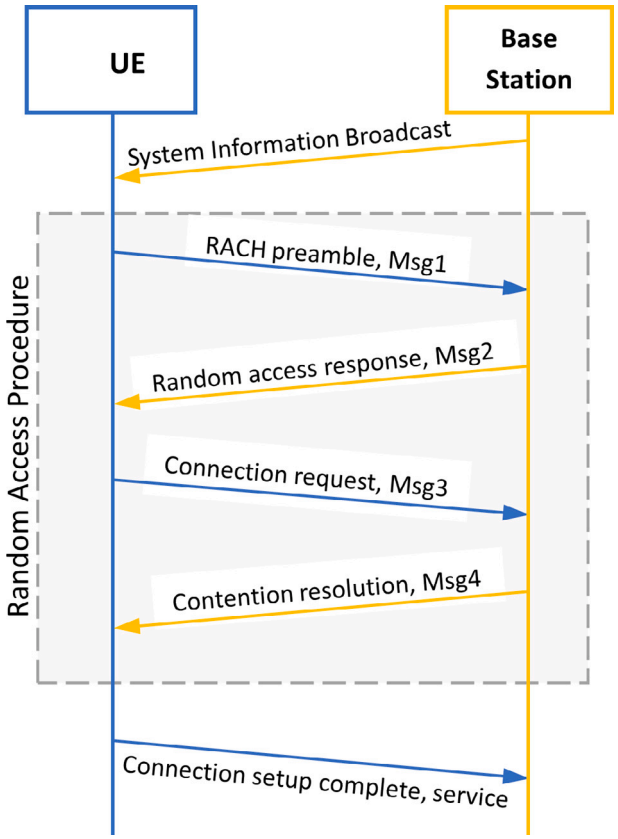


Fig. 1. Contention-based random access procedure.

1. **Preamble Transmission (*Msg1*)**: As a first step of the random access procedure, the UE requests the base station to get an uplink resource, Physical Uplink Shared Channel (PUSCH), for Msg3 transmission. To make the request, the UE randomly selects a preamble from a set of preambles in the cell and sends it to the base station in the upcoming RAO. Of the 64 preambles per cell, 54 of them are used for contention-based mode and the rest are used for contention-free mode. If a preamble transmitted by a UE reaches the base station with sufficient power, it will be detected by the base station, as recommended by 3GPP [28] and verified in [11]. However, if the same preamble is transmitted by more than one UE in the same RAO, preamble collision will occur and hence the base station will not detect any of the UEs preamble transmission [11]. (There is a possibility of preamble detection in such a case [11], but we do not consider it in our work.) Preamble transmission failure occurs for each UE and therefore each UE has try again after a backoff time.
2. **Random Access Response (RAR) (*Msg2*)**: In response to access requests from the UEs, the base station down links Random Access Response (RAR) message (*Msg2*), through Physical Downlink Control Channel (PDCC), to them. The RAR message contains IDs of the detected preambles, unlink grants (PUSCH resources), Backoff Indicator (BI), alignment time, etc. Each UE receives the RAR message and extracts the information relevant to it. Thus, a UE with preamble collision will extract the BI and wait for a random time determined by the BI before trying another preamble transmission.
3. **Connection Request (*Msg3*)**: A UE which has received an uplink grant adjusts its uplink time based on the received alignment time and transmits its connection request message, *Msg3*, through the allocated PUSCH.

4. **Contention Resolution (*Msg4*)**: For a successfully decoded Msg3, the base station responds with a contention resolution message, Msg4. If a UE does not receive a contention resolution message within the contention resolution timer [11], it declares a contention resolution failure and repeats the random access procedure all over again after a backoff time in the next RAO.

If the random access attempt of an UE fails, regardless of the cause (collision or contention resolution timer expiration or insufficient transmission power), the UE repeats the random access procedure all over again [29]. To do this, the UE should first undergo a *backoff procedure*, where the UE waits for a random backoff time, T_{BO} [ms], given below, before trying to transmit a new preamble in the upcoming RAO.

$$T_{BO} = \mathcal{U}(0, BI), \tag{1}$$

where $\mathcal{U}(.)$ stands for uniform distribution, BI is the backoff indicator defined by the base station, $0 \leq \text{BI} \leq 960$ ms. As stated above, every UE that does not get an uplink grant (PUSCH), receives the BI in the RAR message. However, the number of preamble retransmissions is limited to a value called *preambleTransMax*, which is the maximum allowed number of preamble transmissions by a UE in a RAO and it is broadcasted by the base station through the System Information Block Type 2 (SIB2) [23]. In [11], it is shown that $preambleTransMax = 10$ and hence when this limit is exceeded, the network is declared unavailable by the UE and the problem is reported to the upper layer.

The *two-step* RA procedure is included in the Release-16 of the 5G NR standard, and therefore both the four-step and two-step random access procedures are defined in the standard [25–27,30]. The two-step random access procedure reduces the original two-round trip cycles to a single round-trip cycle exchanged between the UE and the base station with the aim of reducing access delay [25,26]. First, the UE sends *MsgA*, which is a combination of *Msg1* and *Msg3*, to the base station. The base station then responds to the UE with *MsgB*, which is a combination of *Msg2* and *Msg4*. *Msg4* may carry either success (success RAR) or failure (fallback RAR).

Though the two-step random access procedure minimizes the latency of the random access procedure, the four-step random access procedure uses more interactions and hence has higher reliability. Therefore, our proposed mechanism can exploit this feature of the four-step random access procedure when integrated into the 5G NR network. In addition, the dominant use of the four-step random access procedure continues. Hence, the four-step random access procedure is relevant for existing (LTE-A and 5G NR) and emerging network standards. For these reasons, we used the four-step random access procedure in our mechanism implementation. This makes our mechanism applicable in both LTE-A and 5G NR network-based mMTC access control.

3.1.1. Access class barring scheme

Access class barring is a method used to redistribute the access requests of UEs in time by limiting the number of access requests per RAO. This reduces the pressure on the RACH pressure. The base station broadcasts a barring factor $P_{ACB} \in [0, 1]$ and mean barring time T_{ACB}, [11], as part of the system information before each RAO. After receiving the barring parameters, the UE generates a random number $q \in [0, 1]$ with a uniform distribution. Then, the UE checks whether $q \leq P_{ACB}$ or not. If $q \leq P_{ACB}$, then the UE randomly selects and sends a preamble to the base station. Otherwise, the UE remains silent and waits for access barring time $T_{barring}$ given by [11],

$$T_{barring} = [0.7 + 0.6 * rand] * T_{ACB}, \tag{2}$$

where $rand = \mathcal{U}[0, 1)$, a uniform distribution random number between 0 and 1. This process is repeated until the UE generates a value q less than P_{ACB} and sends a preamble. Once the UE selects and sends a preamble, the UE goes through steps 2 to 4 of the random access procedure explained above. If there is no contention resolution, the UE goes through backoff procedure and repeats the above mentioned steps until it gets connected. The algorithm of the ACB scheme is given in Algorithm 1.

Algorithm 1: Access Class Barring Scheme

```
repeat
    Set the barring factor P_ACB_i and barring time T_ACB_i
      broadcast by the base station in the i^th SIB2;
    Generate a random number q ∈ [0, 1] with uniform
      distribution;
    if q ≤ P_ACB_i then
        Send a randomly selected preamble;
    else
        Generate another q;
        Assign the barring time
        T_barring = [0.7 + 0.6 * U[0, 1)] * T_ACB_i;
        Wait for T_barring;
until preamble is sent;
```

The details of the random access procedure can be found in Section 2 of our previous work [31]. Additional materials can also be found in [11,12,14].

4. Deep reinforcement learning-based ACB mechanism

This section first gives details of our system model and then the implementation of the proposed mechanism.

4.1. System model

This subsection describes the analysis of the system implementation. The cellular network model, the RACH parameters, and the RL algorithm definition (agent, state, action, reward) are also included.

We followed the 3GPP Release 14 contention based random access procedure specifications in our mechanism implementation. We consider a base station supporting N_M M2M and N_H H2H UEs per cell. This helps to show the impact of M2M UEs traffic on H2H UEs access. Regarding the M2M UEs arrival traffic model, we consider extreme scenario where large number of UEs arrive synchronously following a Beta distribution $p(t)$ over an activation period T, as recommended in 3GPP TR 37.868 [28], and given by,

$$p(t) = \frac{t^{\alpha-1}(T-t)^{\beta-1}}{T^{\alpha+\beta-1}\,Beta(\alpha-\beta)},\ 0 \leq t \leq T, \tag{3}$$

$\alpha > 0, \beta > 0, Beta(\alpha, \beta)$ is the Beta distribution function. The values of $\alpha = 3$ and $\beta = 4$ are used as assumed in [28]. Whereas, the traffic model for the H2H UEs arrival pattern is considered to be uniform over the activation period.

We also take the LTE-A PRACH configuration index to be 6, i.e., *prach-ConfigIndex 6* as specified in [28,29], where the subframe length is 1 ms and the periodicity of the RAO is 5 ms. For *prach-ConfigIndex 6*, system information (including P_{ACB}, T_{ACB}, *prach-ConfigIndex*, *preambleTransMax*, length of the RAR window W_{RAR} (in subframes)) is broadcasted by the base station through SIB2 message every $T_{SIB2} = 16$ RAOs (i.e., 80 ms) [23]. In addition, out of the available 64 preambles per cell, $R = 54$ of them are used for the contention-based random access procedure [11]. And the maximum allowed number of preamble transmissions per RAO for a UE, *preambleTransMax*, is set to be 10 [11]. Additional PRACH parameters for LTE-A Release 14, which are used in our implementations, are listed in Table 1.

Our problem is RACH congestion in a cellular network where there is sporadic massive M2M UEs access request to the base station, in addition to the regular H2H UEs. As a result, the problem can best be solved using model-free reinforcement learning algorithm, as it can learn the

Table 1
PRACH Configuration.

Parameter	Setting
Subcarrier Spacing	15 kHz
PRACH Configuration Index	*prach-ConfigIndex* = 6
Periodicity of RAOs	5 ms
Subframe length	1 ms
Available preambles for contention-based random access	$R = 54$
Maximum number of preamble transmissions	*preambleTransMax* = *10*
RAR window size	$W_{RAR} = 5$ subframes
Maximum number of uplink grants per subframe	$N_{RAR} = 3$
Maximum number of uplink grants per RAR window	$N_{UL} = W_{RAR} \times N_{RAR} = 15$
Preamble detection probability for the kth preamble transmission	$P_d = 1 - \frac{1}{e^k}$ [28]
Backoff Indicator	$BI = 20$ ms
Re-transmission probability for *Msg3* and *Msg4*	0.1
Maximum number of *Msg3* and *Msg4* transmissions	5
Preamble processing delay	2 subframes
Uplink grant processing delay	5 subframes
Connection request processing delay	4 subframes
Round-Trip Time (RTT) of Msg3	8 subframes
RTT of Msg4	5 subframes

changes in the environment through continual interaction with the environment and find an optimal policy that can control the congestion. The optimal policy adjusts the barring factor in response to the access request traffic change in such a way that the probability of preamble collision is minimized or the access success probability is maximized ($P_s \geq 0.95$). And we know that RL problems can be modeled as a Markov Decision Process (MDP) which has five components, namely agent, environment, state, action, and reward. In our case, the *agent* is the *base station*, which interacts with the environment by taking an action and observing the environment's feedback in the form of reward and next state to control the barring factor. And the environment consists of a large number of M2M UEs, with synchronized access request, and regular H2H UEs. The state, action, and reward will be explained in the following paragraph. We use state–action value functions to find the optimal policy that dynamically adjusts the barring factor according to access request traffic to control RACH congestion. The value function can be represented by a *lookup table* or approximated using *neural networks*. In this article, deep neural network representation is used, as detailed in Section 4.

In LTE-A or 5G NR, the base station does not know the number of UEs which are competing for resources nor their access time, which is the case in several centralized access control systems. The base station is able to identify the number of successfully received preambles N_{sp} in a given RAO. Nonetheless, due to factors such as preamble collisions, interference, decoding errors, and other impairments, N_{sp} generally deviates from the total number of transmitted preambles, particularly under highly congested situation. For a base station which uses the dueling double deep Q-learning based ACB mechanism as its access control method, the selection of the barring factor P_{ACB} will be the action taken by the agent to control the RACH congestion. The barring factor P_{ACB} takes one of the 16 possible values in the set $P\{0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4, 0.5, 0.6, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 1\}$. Hence, we define the set of actions as $\mathcal{A} = \{1, 2, 3, \ldots, 16\}$. We also define the state s at any time-step as $s = (P_{ACB}, E[N_{sp}], \Delta E[N_{sp}], s_{N_{sp}})$, where the remaining three of the state features are defined as follows,

- $E[N_{sp}]$ is the mean number of successfully received preambles per RAO and is calculated over the SIB2 message transmission interval, T_{SIB2}, as,

$$E[N_{sp}] = \sum_{j=1}^{T_{SIB2}} N_{sp_j}, \tag{4}$$

- $\Delta E[N_{sp}]$ is the difference between the mean number of successfully received preambles per RAO in the current time-step i and previous time-step $i-1$, i.e.,

$$\Delta E[N_{sp}] = E[N_{sp}]_i - E[N_{sp}]_{i-1}, \tag{5}$$

- $s_{N_{sp}}$ is the normalized sample standard deviation of the number of successfully received preambles, N_{sp}, over T_{SIB2}, one-time step, which is given by

$$s_{N_{sp}} = \frac{\left(\frac{\sum_{j=1}^{T_{SIB2}} (N_{sp} - E[N_{sp}])^2}{T_{SIB2}-1}\right)^{\frac{1}{2}}}{E[N_{sp}]} \tag{6}$$

When the agent takes the action $a = 1$, P_{ACB} changes to 0.05 (i.e., $P_{ACB} = 0.05$), when $a = 2$, $P_{ACB} = 0.1$, when $a = 3$, $P_{ACB} = 0.15$, ..., when $a = 16$, $P_{ACB} = 1$, the ACB scheme will be turned off in this case. And the rest of the state features change as per the P_{ACB} value, where low values of P_{ACB} restrict large number of UEs from transmitting preamble and to the contrary large values of P_{ACB} restrict small number of UEs. Accordingly, changes in the remaining state features can be interpreted. As changes on P_{ACB} can only be received by the UEs through SIB2 message, which is broadcasted every T_{SIB2}, the action that change P_{ACB} and hence other state features will only be taken just before the transmission of SIB2 message. Hence, Fig. 2 shows the state and time-step definitions.

It is important to note that the maximum number of UEs that can successfully access the base station within a single RAO is limited to 54, corresponding to the number of preambles available at the base station for contention-based random access. However, in practice, the actual number of successfully decoded preambles is considerably lower due to connectivity impairments and the likelihood of multiple UEs selecting the same preamble. Our experiments show this fact and hence for any number of competing UEs, the probability of receiving more than 29 successfully decoded preambles per RAO is low. Hence, the possible values of $E[N_{sp}]$ are taken to be between 0 and 29. To limit the number of states, we discretized $s_{N_{sp}}$ such that $s_{N_{sp}} \in \{0, 0.2, 0.4, 0.6, 0.8\}$. That is, over a time-step i, if $0 \leq s_{N_{sp}} < 0.2$, we take $s_{N_{sp}} = 0$; if $0.2 \leq s_{N_{sp}} < 0.4$, we take $s_{N_{sp}} = 0.2$, and similarly for the other intervals. In addition, $\Delta E[N_{sp}]$ is discretized as follows: if $\Delta E[N_{sp}] > 0$, then it will be taken as 1; if $\Delta E[N_{sp}] < 0$, then it will be taken as 2; if $\Delta E[N_{sp}] = 0$, then it will be taken as 3. Finally, P_{ACB_i} is the barring factor in the current time-step i.

4.2. Dueling double deep Q-learning-based ACB mechanism

This subsection explains the implementation of the proposed reinforcement learning-based ACB mechanism. As the mechanism is based on dueling double deep Q-learning algorithm, its explanation is included in this section. Then, the algorithm of the proposed mechanism follows.

In standard Q-learning, the Q-value in a state s and current action a is updated using the following formula [32]

$$Q(s,a) = Q(s,a) + \alpha\big(r + \gamma \max_a Q(s',a) - Q(s,a)\big), \tag{7}$$

where $\alpha \in (0,1]$ is the learning rate, $\gamma \in [0,1]$ is a discount factor that trades-off the significance of immediate and future rewards, r is the immediate reward for taking the action in state s. The action in the next state s' corresponds to the maximum Q-value in the state. Such a selection of the action results in overestimation bias. To minimize the overestimation bias and other features of Q-learning, a number of

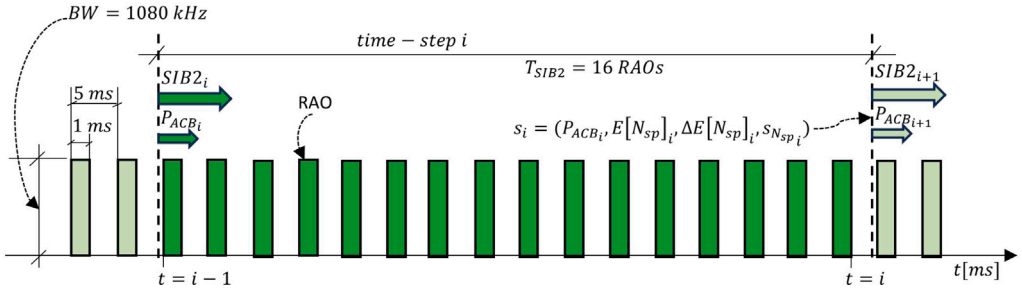


Fig. 2. State and time-step definitions.

modifications have been introduced into the standard Q-learning [33–36].

As it is explained below, the dueling double deep Q-learning algorithm estimates the Q-value as a combination of state-value function (measures state importance) and advantage function (measures action importance). This gives the algorithm a faster learning speed than Double Deep Q-Learning (DDQL) as it can learn which states matter though actions may have similar Q-values. It also has the strongest generalization ability compared with DDQL, Deep Q-Learning (DQL), and classical RL algorithms for its separated state-value and action advantage. In addition, it has a reduced overestimation bias which is inherited from its base algorithm DDQL that uses two deep Q networks (online and target) for this purpose [36].

Hence, for our proposed mechanism, we prefer to use the Dueling DDQL algorithm to dynamically adjust the barring factor parameter in response to the changing UEs traffic. The Dueling DDQL algorithm approximates the state–action value function using Dueling Deep Q-Network (Dueling DQN). In addition, the algorithm utilizes two Dueling DQNs namely, *online* and *target* Dueling DQN. The online network is used to select the action a_{max} that gives maximum Q-value in the next state s' and the target network is used to evaluate this action. The implemented (online) Dueling DQN architecture is shown in Fig. 3. It has four inputs, one for each state feature, and sixteen outputs, one for each action. The network uses two streams of fully connected layers to estimate the Q-value [36], in contrast to Double Deep Q-Network (DDQN), which utilizes a single stream of fully connected layers for the same purpose [34]. The upper stream outputs the value of a state $V(s;\theta,\theta_V)$, a scalar which measures the importance of a state and given by

$$V(s;\theta,\theta_V) = E\big[Q(s,a;\theta,\theta_V,\theta_A)\big], \tag{8}$$

and the lower stream outputs the advantage of each action in a state, $A(s,a;\theta;\theta_A)$, $|\mathcal{A}|$-dimensional vector given by [36],

$$A(s,a;\theta;\theta_A) = Q(s,a;\theta,\theta_V,\theta_A) - V(s;\theta,\theta_V). \tag{9}$$

Note that θ represents the parameters of the common layers, whereas θ_V and θ_A are the parameters of the upper and lower streams, respectively. Then, the two streams are combined to give the Q-values in a state s, which is $|\mathcal{A}|$-dimensional vector [36]

$$Q(s,a;\theta,\theta_V,\theta_A) = V(s;\theta,\theta_V) + \Big(A(s,a;\theta,\theta_A) - \frac{1}{\mathcal{A}}\sum_{a'} A(s,a';\theta,\theta_A)\Big). \tag{10}$$

The action that maximizes the Q-value in the next state s' is then,

$$a_{max} = \arg\max_{a'} Q(s',a';\theta,\theta_V,\theta_A) \tag{11}$$

The target network, $Q(s,a;\theta',\theta'_V,\theta'_A)$, has the same parameters as the online network, $Q(s,a;\theta,\theta_V,\theta_A)$, i.e., $\theta' = \theta, \theta'_V = \theta_V, \theta'_A = \theta_A$.

Using Eq. (11), the target equation is then given by [36]

$$Y = r + \gamma Q(s', a_{max};\theta',\theta'_V,\theta'_A). \tag{12}$$

To learn the optimal Q-values, we optimize the following sequence of loss function at iteration i [36]

$$L_i(\theta_i,\theta_{V_i},\theta_{A_i}) = E_{s,a,r,s'}\big[\big(Y_i - Q(s,a;\theta_i,\theta_{V_i},\theta_{A_i})\big)^2\big]. \tag{13}$$

The best way to learn the optimal parameters of the two networks is to freeze the parameters of the target network, $Q(s,a;\theta',\theta'_V,\theta'_A)$, for a fixed number of iterations τ while updating the online network, $Q(s,a;\theta,\theta_V,\theta_A)$, by gradient descent [35]. The gradient update is given by [36]

$$\nabla_{\theta_i\theta_{V_i}\theta_{A_i}} L_i(\theta_i,\theta_{V_i},\theta_{A_i}) = E_{s,a,r,s'}\big[\big(Y - Q(s,a;\theta_i,\theta_{V_i},\theta_{A_i})\big)\nabla_{\theta_i\theta_{V_i}\theta_{A_i}} Q(s,a;\theta_i,\theta_{V_i},\theta_{A_i})\big]. \tag{14}$$

Then, the target network parameters are copied from the online network parameters so that $\theta' = \theta, \theta'_V = \theta_V, \theta'_A = \theta_A$.

The Dueling DDQL algorithm has one other important component called *Experience Replay (ER)* [35]. During learning the agent stores its transition experiences $e_t = (s_t, a_t, r_t, s_{t+1})$ from many episodes as a data set $\mathcal{D}_t = \{e_1, e_2, \ldots, e_t\}$. When the Dueling DDQN is trained, instead of only using the current experience as prescribed by the standard Q-learning [32,36], the network is trained by sampling mini-batches of experiences from D uniformly at random. Accordingly, the sequence of loss function will be [36]

$$L_i(\theta_i,\theta_{V_i},\theta_{A_i}) = E_{(s,a,r,s')\sim\mathcal{U}(D)}\big[\big(Y_i - Q(s,a;\theta_i,\theta_{V_i},\theta_{A_i})\big)^2\big]. \tag{15}$$

Experience replay increases data efficiency through re-use of experience samples in several updates. Moreover, it reduces variance because uniform sampling reduces the correlation among the samples used in the update.

The algorithm for Dueling DDQL-Based ACB Mechanism is given in Algorithm 2. To avoid possible confusion, the online and target networks are denoted by $Q_O(s,a;\theta,\theta_V,\theta_A)$ and $Q_T(s,a;\theta',\theta'_V,\theta'_A)$ respectively.

5. Results and discussions

This Section presents the performance evaluation results of our implemented mechanism evaluated using 3GPP's RACH Key Performance Indicators (KPIs), namely the probability of successful access (P_s), the number of preamble transmissions (K) for the successfully connected UEs, and the access delay (D).

We considered one of the cells of the base station which supports both M2M and H2H UEs with different access request intensities. To make our evaluation more realistic, for the H2H UEs arrival distribution, we use the Open Dataset, which is released in Open Data in December 2014, provided by the Italian operator, Telecom Italia [37]. The dataset was initially (January 2014) used in a contest (entitled "The Telecom Italia Big Data Challenge") for innovative application development from Big Data with the release of one the largest heterogeneous Big Data set. The data set provides information about the telecommunications activity (voice, data, SMS) over the cities of Milano and Trento. The dataset is the result of a computation over the Call Detail Records (CDRs) generated by the operator's cellular network over the two cities. The dataset has spatial and temporal aggregation values. The temporal values are aggregated in periods of ten minutes over the months of November and December 2013. Hence, the data is assumed to be constant over every 10 min period. As stated in [38],

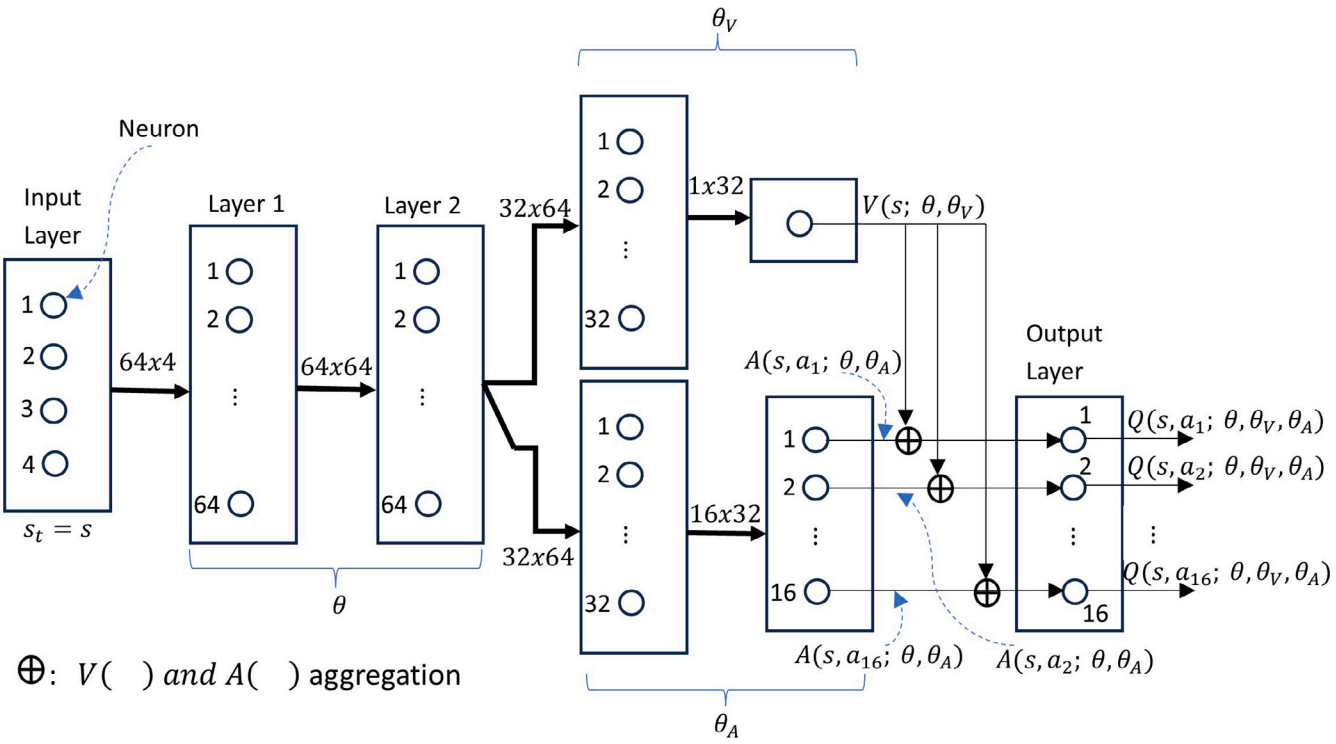


Fig. 3. Implemented Dueling Deep Q-Network architecture.

the base station can support a maximum of 55 Radio Access Bearer (RAB) setups per second and hence we use this capacity as the threshold capacity of the base station. Accordingly, the aggregated H2H data traffic is normalized by this threshold value. Regarding the M2M UEs arrival pattern, it follows a $\beta(3,4)$ distribution over a period of $T = 10$ seconds as stated in Section 4. Hence, in our simulations, the M2M UEs traffic is added to the H2H traffic in every 10 min period to simulate a massive access scenario. Note that in generating the beta-distributed M2M traffic, we used a unique seed for each 10 min interval. The number of M2M UEs used in our experiments (using Dell Workstation Computer (Processor: Intel(R) Xeon(R) Gold 6230 CPU @ 2.10 GHz 2.10 GHz, RAM: 32.0 GB)) are 10,000, 30,000, and 60,000. For the H2H UEs traffic we used November 2 data (the highest traffic load in the two month record) in training and December 1 data (the second highest traffic load in the period) in testing the mechanism. Both data were taken from cell 5161, one of the most occupied cells during the two months. Section 5.1 gives the results of the experiments we conducted to select the learning rate and discount factor values and found that $\alpha = 0.15$ and $\gamma = 0.7$, are good choices. As we used unique seed for every 10 min period (of M2M UEs traffic generation), we are able to conduct multiple (200 and 100) different tests and hence the result shown for a specific parameter is the average of the parameter taken from the multiple test results.

5.1. Dueling double deep Q-learning-based ACB mechanism

Each of the following results is an average of 200 different experiments, each conducted over a day, for a loaded scenario containing the traffic of 30,000 M2M UEs and H2H UEs in every 10 min.

It is shown in [11] that an access congestion control mechanism should maintain the number of UEs arrivals per RAO, N, below the number of uplink grants that the base station can send between two consecutive RAOs, N_{UL}, and the PRACH capacity, $c(R)$, i.e., $N \leq min\{N_{UL}, c(R)\}$. For $R = 54$, $c(R)$ is found to be 20.05 [11], and $N_{UL} = 15$ as it is given in Table 1. Therefore, $N \leq min\{15, 20.05\}$.

Different plots of our mechanism's performance evaluation results are shown below from Figs. 4 to 6. As can be seen in Fig. 4, the peak value of the average number of total preamble transmissions (first time preamble transmissions plus backlogged UEs preamble transmissions) per RAO, which is attained at the 1000th RAO (5 s from the start of activation of M2M UEs), is less than 20, which is in turn less than the RACH capacity, $c(54) = 20.05$ [11]. This shows that the mechanism is limiting the number of UEs requesting for network access to the RACH capacity. In Fig. 5, it can be observed that the barring factor is reduced to its minimum value of around 0.27 at the 1000th RAO. This indicates that the mechanism is adaptively adjusting the barring factor, i.e., it decreases the barring factor to its lowest value over the activation period when the number UEs requesting for access reaches its peak value in the period. As a result, the mechanism is working as per the design requirement. To give another illustrative example showing the mechanism's functionality as required, in Fig. 4, at the 5000th RAO, the average number of total preamble transmissions per RAO is around 3, which is less than the $min\{N_{UL}, c(R)\}$. And in Fig. 5, the value of the barring factor at this time ($5,000^{th}$ RAO) is around 0.9, which lets large number of arriving UEs for access request. As it can be noted in Fig. 5, from the $6,000^{th}$ RAO on wards, the barring factor is 1, i.e., ACB is turned-off as the activation period of the M2M UEs is over; in other words, the extreme access request scenario by the massive arrival of the M2M UEs is limited to the activation period and once that period is over, the base station serves the regular H2H UEs access requests and $P_{ACB} = 1$ lets the H2H UEs get access without congestion. From Fig. 4, it can also be observed that the peak value of average number of first preamble transmissions per RAO is less than 10, which is less than $min\{N_{UL}, c(R)\}$ as per the congestion control mechanism requirement. In addition, the average number of successful access per RAO almost coincides with the average number of first preamble transmissions per RAO throughout the activation period indicating that almost every UE is getting an uplink grant in it first preamble transmission. Moreover, the average number of collisions per RAO is significantly less than the successful access per RAO throughout the activation period which is as required for a properly functioning congestion control mechanism. This indicates that preamble collision is considerably minimized.

In Fig. 6, the peak value of the average number of successful preambles per RAO is around 12.5, which is less than the $N_{UL} = 15$

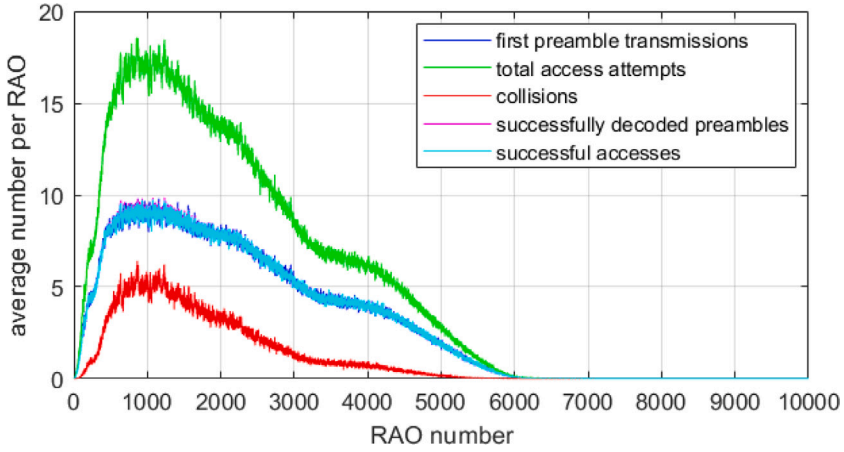


Fig. 4. UEs first preamble transmissions, total preamble transmissions, number of collisions, successfully decoded preambles, and successful access per RAO.

Algorithm 2: Dueling Double Deep QL-Based ACB Mechanism

Controller: Dueling Double Deep QL($\mathcal{S}$, $\mathcal{A}$, $\mathcal{R}$, γ, ϵ, τ, ER, N_l)
Input: set of states $\mathcal{S}$, set of actions $\mathcal{A}$, reward function $\mathcal{R}$, discount factor γ, exploration probability ϵ, target update frequency τ, experience replay size ER, neural network layers N_l
Local Variables:
Dueling DQN $Q_O(s,a;\theta,\theta_V,\theta_A)$, state s, action a, weights θ, θ_V, θ_A
Dueling DQN $Q_T(s,a;\theta',\theta_V',\theta_A')$, state s, action a, weights θ', θ_V', θ_A')
Initialization:
Initialize $Q_O(s,a;\theta,\theta_V,\theta_A)$ with random weights θ,θ_V,θ_A
Initialize $Q_T(s,a;\theta',\theta_V',\theta_A')$ i.e., $\theta'=\theta,\ \theta_V'=\theta_V,\ \theta_A'=\theta_A$
Initialize the experience reply buffer
repeat
  repeat
    if $RAO(i) \mod T_{SIB2} == 0$ **then**
      Observe current state s;
      Select action a from $\mathcal{A}$ using ϵ-greedy policy based on $Q_O(s,a;\theta,\theta_V,\theta_A)$;
      Execute action a, observe reward $r(s,a,s')$ & next state s';
      STORE (s,a,r,s') in the experience replay buffer;
  until $j = ER$;
  $s = s'$;
  Randomly order the buffer;
  $l = 0$;
  repeat
    $n = l\tau + 1$;
    repeat
      $a_{max} = \arg\max_{a\in A} Q_O(s,a;\theta,\theta_V,\theta_A)$;
      $target = r + \gamma Q_T(s',a_{max};\theta',\theta_V',\theta_A')$;
      STORE *target* Q-values in buffer;
    until $n = (l+1)\tau$;
    Read the buffer for s and target (τ each);
    Compute loss and gradient;
    $\mathcal{L} = \frac{1}{\tau}\sum_n (target - Q_O(s,a;\theta,\theta_V,\theta_A))^2$;
    Train $Q_O(s,a;\theta,\theta_V,\theta_A)$;
    $Q_T(s,a;\theta',\theta_V',\theta_A') = Q_O(s,a;\theta,\theta_V,\theta_A)$;
  until $l = ER/\tau$;
until $i = maxRAO$;

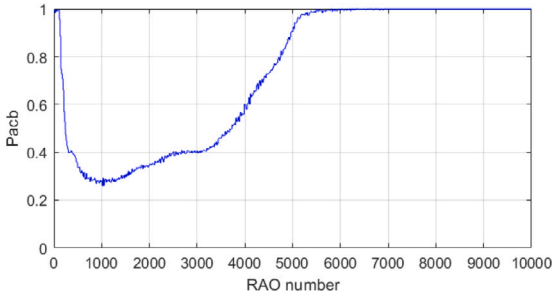


Fig. 5. The barring factor, P_{ACB}, adaptation as a function of time.

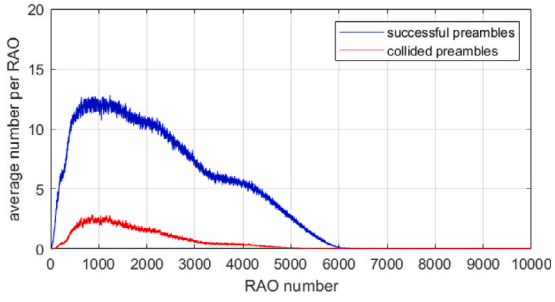


Fig. 6. Successful preambles, collided preambles per RAO.

indicating that the mechanism minimizes preamble collisions by not reaching the red cap. And the peak value of the average number of collided preambles per RAO is around 2.49, a difference of 10.01 from the peak value of average number of successful preambles per RAO. Therefore, the mechanism very significantly reduces preamble collision.

Tables 2 and 3 below show the effects of the learning rate α and the discount factor γ on the performance of the mechanism. Table 2 shows KPI values for $\gamma = 0.7$ and three learning rate values (from its range $\alpha \in (0,1]$) for M2M, H2H, and all UE types. The learning rate has no significant effect on the KPIs of the mechanism and hence we take $\alpha = 0.15$ in our simulations as it gives relatively better access success probability and access delay performances. Similarly, Table 3 shows the KPI values for $\alpha = 0.15$ and three discount factor values (from its range $\gamma \in [0,1]$). For $\gamma = 0.95$, the access success probability and number of preamble transmission performances are poor. Hence, future rewards negatively affect the mechanism's performance and hence reasonable not to take large γ values. The access success probability and the number of preamble transmission performances are better for $\gamma = 0.7$ and used in our simulations.

Table 2
KPIs for the Dueling DDQL Mechanism as α varies for $N_m = 30,000$ and $\gamma = 0.7$.

	Access Success Prob.			Mean Access Delay [ms]			Mean No. Preamb. TxN		
	$\alpha = 0.02$	$\alpha = 0.15$	$\alpha = 0.28$	$\alpha = 0.02$	$\alpha = 0.15$	$\alpha = 0.28$	$\alpha = 0.02$	$\alpha = 0.15$	$\alpha = 0.28$
M2M	0.9999	0.9998	0.9998	7351.34	6739.83	7106.63	1.6914	1.7195	1.6978
H2H	1.0000	1.0000	1.0000	3703.81	3288.72	3527.94	1.5646	1.5760	1.5691
All UEs	0.9999	0.9998	0.9998	7340.43	6730.05	7096.39	1.6910	1.7191	1.6974

Table 3
KPIs for the Dueling DDQL Mechanism as γ varies for $N_m = 30,000$ and $\alpha = 0.15$.

	Access Success Prob.			Mean Access Delay [ms]			Mean No. Preamb. TxN		
	$\gamma = 0.2$	$\gamma = 0.7$	$\gamma = 0.95$	$\gamma = 0.2$	$\gamma = 0.7$	$\gamma = 0.95$	$\gamma = 0.2$	$\gamma = 0.7$	$\gamma = 0.95$
M2M	0.9999	0.9998	0.7878	6539.29	6739.83	2967.51	1.7300	1.7195	2.5444
H2H	1.0000	1.0000	0.9363	3165.82	3288.72	1243.65	1.5591	1.5760	1.7212
All UEs	0.9999	0.9998	0.7881	6529.40	6730.05	2962.34	1.7295	1.7191	2.5414

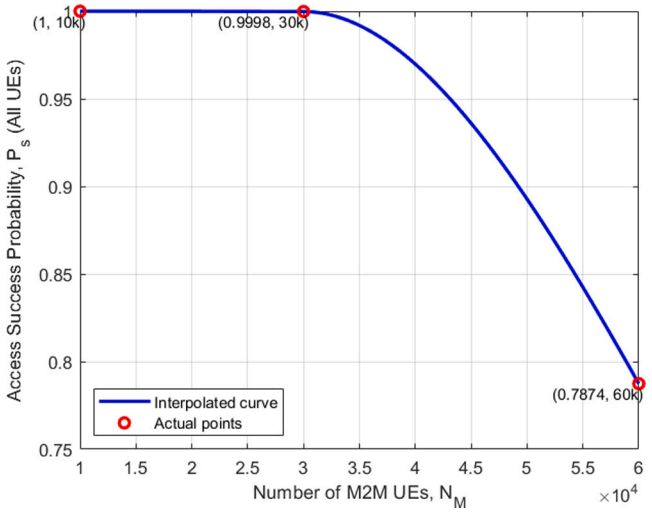


Fig. 7. Access success probability for all UEs.

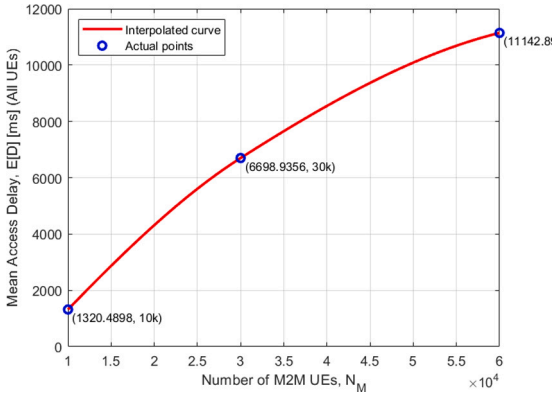


Fig. 8. Mean access delay for all UEs.

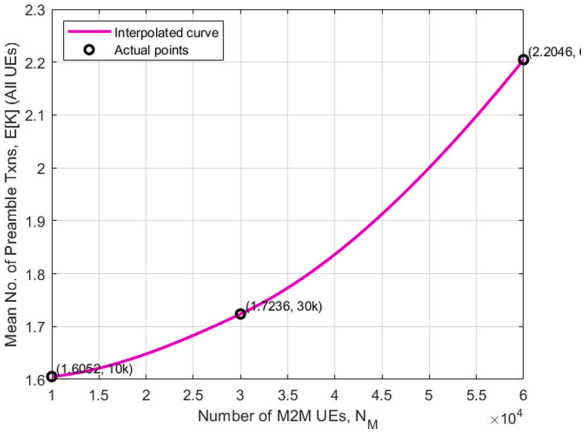


Fig. 9. Mean number of preamble transmissions for all UEs.

We evaluated the performance of our mechanism as the number of M2M UEs increases. Accordingly, under the access control of the mechanism, the access success probability (P_s) of all UEs (M2M + H2H) is measured against the number of M2M UEs (for $N_M = 10,000, 30,000, 60,000$). A plot of N_M vs P_s is shown in Fig. 7. The result shows that the access success probability performance of the mechanism meets the optimal requirement ($P_s \geq 0.95$), with a maximum of 0.9998, for up to 40,250 M2M UEs. (That is, more than 95% of the UEs that requested for access will get successful access to the base station.) That means, the access control performance of the mechanism is optimum for up to 40,250 M2M UEs (in the presence of the H2H UEs). The access control performance of the mechanism is still good for up to 60,000 M2M UEs, with $P_s \geq 0.7874f$. Furthermore, Figs. 8 and 9 show the mean access delay and number of preamble transmission variations with the number of M2M UEs for the mechanism. Unlike the mean number of preamble transmissions, the mean access delay becomes asymptotic to a maximum value as the number of M2M UEs increases. Depending on the specific application, tolerable access delay and number of preamble transmission (energy consumption) can be selected for up to 40,250 M2M UEs.

5.2. Comparison of the dueling double deep Q-learning-based ACB mechanism and RL-based ACB scheme

Results of comparative plots of our Dueling Double Deep Q-Learning-Based ACB Mechanism and the RL-Based ACB Scheme implemented in [12] are shown from Figs. 10 to 12. The simulations are done for 30,000 M2M UEs in the presence of H2H UEs. Fig. 10 shows that both mechanisms perform almost similarly in terms of average number of total preamble transmissions, first preamble transmissions, successful access, and preamble collision per RAO. The slight difference is due to the Q-Learning algorithm's overestimation problem. Fig. 11 shows that both control mechanisms adjust P_{ACB} almost similarly. Fig. 12 also shows that in terms of average number of successful preambles and collided preambles per RAO, both mechanisms perform almost similarly.

Fig. 13 shows the access success probability performances of the Dueling DDQL-Based ACB Mechanism and the RL-Based ACB Scheme as the number of M2M UEs varies. Each P_s at a key point is an average of 100 test results and the remaining points on a curve are obtained

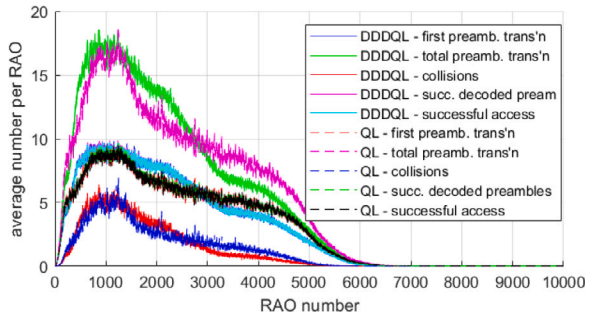


Fig. 10. UEs first preamble transmissions, total preamble transmissions, number of collisions, successfully decoded preambles, and successful access per RAO.

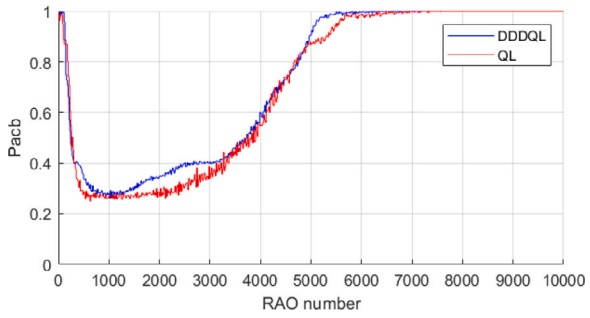


Fig. 11. The barring factor, P_{ACB}, adaptation as a function of time.

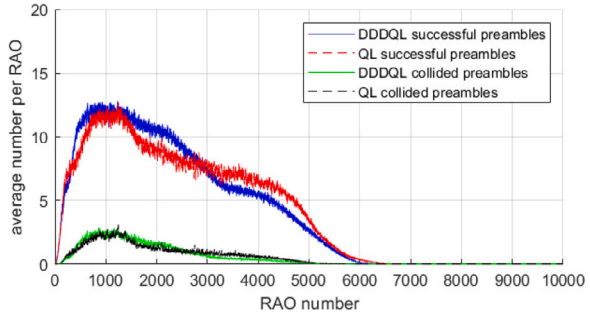


Fig. 12. Successful preambles, collided preambles per RAO.

through interpolation. The result shows that our mechanism achieves optimum performance ($P_s \geq 0.95$) for up to 40,250 M2M UEs and the performance degradation with the number of M2M UEs is much slower than the RL-Based ACB Scheme. The RL-Based ACB Scheme performs optimally for up to 35,000 M2M UEs and performs unacceptably ($P_s = 0.1272$) for 60,000 M2M UEs for which our mechanism still has good performance ($P_s = 0.7446$).

In summary, for $N_m = 30{,}000$ and $N_m = 10{,}000$, the Dueling Double Deep Q-Learning-Based ACB Mechanism and RL-Based ACB Scheme have similar performance in improving the access success probability relative to the static optimal ACB scheme, ($P_{ACB} = 0.5$, $T_{ACB} = 4$ s), performance. As the number of M2M UEs increases beyond 30,000, the access success probability performance of the two mechanisms show degradation where the RL-ACB scheme degrading rapidly. Therefore, based on the results and the generalization ability of Dueling DDQL, we can conclude that our mechanism performs much better than the RL-ACB Scheme under a high traffic load. In [12] it is shown that the RL-Based ACB Scheme reduces the number of preamble collisions much better than the static optimal ACB scheme. In this work, we only compare the performances of our mechanism with the RL-Based ACB

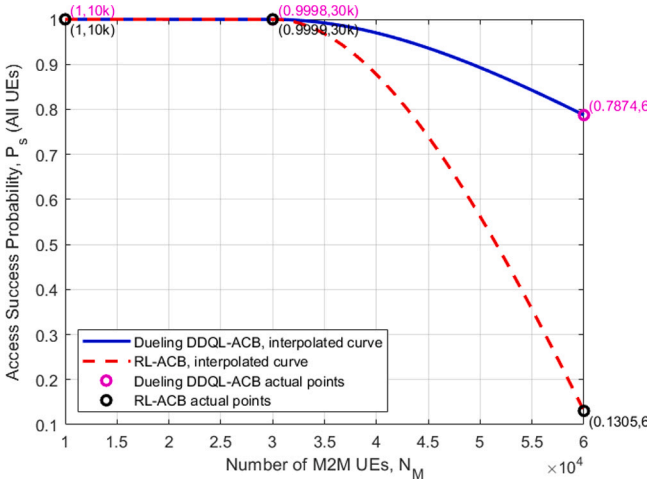


Fig. 13. Access success probability for all UE types.

Table 4
KPIs for the Dueling DDQL-Based ACB Mechanism and RL-Based ACB Scheme for $N_M = 30{,}000$.

KPI	Dueling DDQL-Based ACB Mechanism	RL-Based ACB Scheme
P_s	0.9998	0.9999
P_{sM}	0.9998	0.9999
P_{sH}	1.0000	1.0000
$E[D]$	6698.9356	7601.4898
$E[D]_M$	6708.5782	7614.6221
$E[D]_H$	3273.0683	3405.9426
$E[K]$	1.7236	1.7267
$E[K]_M$	1.7239	1.7272
$E[K]_H$	1.5945	1.5658

Scheme. Hence, when it is required to state our results relative to the static optimal ACB, we use the results in [12].

The RACH KPIs of our proposed method and the reference scheme in [12] are shown in Table 4. It can be seen that, the two methods have similar access success probability performance. However, our mechanism outperforms the reference scheme in terms of mean access delay, making it preferable for delay-sensitive applications. Furthermore, the mean number of preambles transmitted by all UEs (M2M + H2H) and M2M UEs are less than that of the reference scheme, which makes our proposed solution preferable for the energy-limited M2M UEs. Therefore, our mechanism effectively controls the RACH congestion with better overall performance than the reference scheme.

6. Conclusion

In this paper, we proposed a dynamic ACB mechanism which aims to improve the access success probability performance of the ACB scheme. The mechanism is designed for a cellular network in which both M2M and H2H UEs coexist. For the M2M UEs arrival traffic, we used a model recommended by 3GPP. To make the evaluation of our mechanism more realistic, we used records of an operator for the H2H UEs arrival traffic. As we used 3GPP standards in implementation, the mechanism conforms to the specifications of the access procedure in cellular networks (LTE-A and 5G) and therefore it can be seamlessly integrated into the networks. The simulation results show that the proposed mechanism achieves maximum access success probability (0.9998), which meets the theoretical optimum ACB performance ($P_s \geq 0.95$). For number of M2M UEs $N_M \leq 30{,}000$, our mechanism achieves similar access success probability performance (0.9998) as the reference RL-based ACB scheme, but with lower access delay and smaller number of preamble transmissions (energy consumption). Hence, our

proposed method is preferable for the energy-constrained M2M UEs access control. It is also a suitable choice for delay sensitive applications. In addition, under high traffic load ($N_M > 30,000$), our mechanism achieves better access success than the reference scheme.

CRediT authorship contribution statement

Workineh Gebeye Abera: Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Thomas O. Olwal:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Conceptualization. **Yihenew Wondie Marye:** Writing – review & editing, Visualization, Validation, Supervision,.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] IoT Market Analysis. 2024, https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-for-development/wp-content/uploads/2023/04/IoT-for-Development-Use-cases-delivering-impact.pdf.

[2] Global IoT connections forecast to reach 40 billion in 2033. 2024, https://transformainsights.com/news/iot-connections-40-billion-2033.

[3] Akpakwu Godfrey Anuga, Silva Bruno J, Hancke Gerhard P, Abu-Mahfouz Adnan M. A survey on 5G networks for the internet of things: Communication technologies and challenges. IEEE Access 2017;6:3619–47.

[4] Global IoT Forecast Report, 2023–2033. 2024, https://transformainsights.com/research/reports/global-iot-forecast-report-2023-2033.

[5] IoT for Development: use cases delivering impact. 2023, https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-for-development/wp-content/uploads/2023/04/IoT-for-Development-Use-cases-delivering-impact.pdf.

[6] Mobile IoT in the 5G future: NB-IoT and LTE-M in the context of 5G. 2018, https://www.gsma.com/solutions-and-impact/technologies/internet-of-things/wp-content/uploads/2018/05/GSMA-5G-Mobilea-IoT.pdf.

[7] Qadir Qahhar Muhammad, Rashid Tarik A, Al-Salihi Nawzad K, Ismael Birzo, Kist Alexander A, Zhang Zhongwei. Low power wide area networks: A survey of enabling technologies, applications and interoperability needs. IEEE Access 2018;6:77454–73.

[8] Buurman Ben, Kamruzzaman Joarder, Karmakar Gour, Islam Syed. Low-power wide-area networks: Design goals, architecture, suitability to use cases and research challenges. IEEE Access 2020;8:17179–220.

[9] Chilamkurthy Naga Srinivasarao, Pandey Om Jee, Ghosh Anirban, Cenkeramaddi Linga Reddy, Dai Hong Ning. Low-power wide-area networks: A broad overview of its different aspects. IEEE Access 2022;10:17179–220.

[10] LTE-to-5G Evolution January 2024. 2024, https://gsacom.com/paper/lte-to-5G-evolution-january-2024/.

[11] Tello-Oquendo Luis, Leyva-Mayorga Israel, Pla Vicent, Martinez-Bauset Jorge, Vidal Jos'e-Ram'on, Casares-Giner Vicente, Guijarro Luis. Performance analysis and optimal access class barring parameter configuration in LTE-A networks with massive M2M traffic. IEEE Trans Veh Technol 2018;67(4):3505–20.

[12] Tello-Oquendo Luis, Pacheco-Paramo Diego, Pla Vicent, Martinez-Bauset Jorge. Reinforcement learning-based ACB in LTE-A networks for handling massive M2M and H2H communications. In: 2018 IEEE international conference on communications. ICC, Kansas City, MO, USA: IEEE; 2018, p. 1–7.

[13] NR; Physical channels and modulation. 3GPP TS 38.211, Version 18.4.0 Release 18, 2024.

[14] Duan Suyang, Shah-Mansouri Vahid, Wang Zehua, Wong Vincent. D-ACB: Adaptive congestion control algorithm for bursty M2M traffic in LTE networks. IIEEE Trans Veh Technol 2016;65(12):9847–61.

[15] Tello-Oquendo Luis, Vidal José-Ramón, Pla Vicent, Guijarro Luis. Dynamic access class barring parameter tuning in LTE-A networks with massive M2M traffic. In: 2018 17th annual mediterranean ad hoc networking workshop (med-hoc-net). Capri, Italy: IEEE; 2018, p. 1–8.

[16] Moon Jihun, Lim Yujin. Adaptive access class barring for machine-type communications in LTE-A. In: W2016 eighth international conference on ubiquitous and future networks. ICUFN, Vienna: IEEE; 2016, p. 398–402.

[17] Iiyambo Loini, Hancke Gerhard, Abu-Mahfouz Adnan M. A survey on NB-IoT random access: Approaches for uplink radio access network congestion management. IEEE Access 2024;12:95487–506.

[18] Tu Yue, Xu Rongtao. DGPA-ACB: Adaptive RAN overload avoidance control for mMTC in 5G NR. In: 2020 international conference on wireless communications and signal processing. WCSP, Nanjing, China: IEEE; 2020, p. 777–82.

[19] Lee Chu-Heng, Kao Shang-Juh, Chang Fu-Min. LSTM-based ACB scheme for machine type communications in LTE-A networks. Elsevier, Comput Commun 2020;152:296–304.

[20] Llobet Martí, Cabrera-Bean Margarita, Vidal Josep, Agustin Adrian. Optimizing access demand for mMTC traffic using neural networks. IEEE Trans Veh Technol 2023;72(12):16834–8.

[21] Turan Ali, Koseoglu Mehmet, Akcapinar Sezer Ebru. Reinforcement learning based adaptive access class barring for RAN slicing. In: 2021 IEEE international conference on communications workshops (ICC workshops). Montreal, QC, Canada: IEEE; 2021, p. 1–6.

[22] Chen Ziqi, Smith David B. Heterogeneous machine-type communications in cellular networks: Random access optimization by deep reinforcement learning. In: 2018 IEEE international conference on communications. ICC, Kansas City, MO, USA: IEEE; 2018, p. 1–6.

[23] Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification. 3GPP TS 36.331, Version 18.3.1 Release 18, 2024.

[24] Evolved Universal Terrestrial Radio Access (E-UTRA); Physical channels and modulation. 3GPP TS 36.211, Version 18.0.1 Release 18, 2024.

[25] Piao Yijun, Lee Tae-Jin. Integrated 2–4 step random access for heterogeneous and massive IoT devices. IEEE Trans Green Commun Netw 2024;8(1):441–52.

[26] Miuccio Luciano, Panno Daniela, Riolo Salvatore. An energy-efficient DL-aided massive multiple access scheme for IoT scenarios in beyond 5G networks. IEEE Internet Things J 2023;10(9):7936–59.

[27] Liu Xue, Yang Heng, Li Shanshan, Liu Zhenyu, Lian Xiaohui. Enhanced RACH optimization in IoT networks: A DQN approach for balancing H2H and M2M communications. Elsevier Internet Things 2024;28:101433.

[28] RAN improvements for machine-type communications. 3GPP TR 37.868, Version 11.0.0 Release 11, 2011.

[29] Evolved Universal Terrestrial Radio Access (E-UTRA); Medium Access Control (MAC) protocol specification. 3GPP TS 36.321, Version 18.3.0 Release 18, 2024.

[30] Wang Jiadai, Xing Chaochao, Liu Jiajia. Intelligent preamble allocation for coexistence of mMTC/URLLC devices: A hierarchical Q-learning based approach. China Commun 2023;20(8):44–53.

[31] Abera Workineh Gebeye, Olwal Thomas O, Marye Yihenew Wondie, Abebe Abiy Tadesse. Learning based access class barring for massive machine type communication random access congestion control in LTE-A networks. In: 2021 international conference on electrical, computer and energy technologies. ICECET, Cape Town, South Africa: IEEE; 2022, p. 1–7.

[32] Suttun Rechard S, Barto Andrew G. Reinforcement learning: An introduction, In: Adaptive computation and machine learning, 2nd ed.. Cambridge, Massachusetts: The MIT Press; 2020.

[33] Hasselt Hado Van. Double Q-learning. In: 24th annual conference on neural information processing systems. Vancouver, Canada; 2010.

[34] van Hasselt Hado, Guez Arthur, Silver David. Deep reinforcement learning with double Q-Learning. In: The thirtieth AAAI conference on artificial intelligence, vol. 30. Phoenix, USA; 2016.

[35] Mnih Volodymyr, et al. Human-level control through deep reinforcement learning. Nature 2015;518:529–33.

[36] Wang Z, Schaul T, Hessel M, van Hasselt H, de Freitas MLanctot N. Dueling network architectures for deep reinforcement learning. In: The 33rd international conference on machine learning, vol. 48. New York, USA; 2016.

[37] Telecom Italia. The Telecom Italia Big Data Challenge. 2014, https://theodi.fbk.eu/openbigdata/#portfolioModal13.

[38] Nokia. High capacity mobile broadband for mass events. 2016, Nokia Networks white paper.